Lorentzian Geometry from the Causal Preorder

Lifting the Finite Prototype to the Continuum

A Proposed Paper BG for the Dream-Paper Suite

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Abstract

Paper BD proves the Causal Reconstruction Theorem: the causal preorder $\preceq_{\text{caus}}$ emerges from the Lieb–Robinson bound on the transfer operator e^{-tL_∞}, with finite propagation speed $v_{\max} = \sqrt{\|[F_\nabla, \cdot]\|/\mathfrak{g}_\infty}$. The NFC monograph contains Theorem 15.6 (Lorentzian Signature): in the finite prototype, the irreversible ledger direction plus spatial isotropy plus causal cone forces $(v_{\max}, +, +, +)$ Lorentzian signature, with no Minkowski postulates.

The present paper, Paper BG, lifts Theorem 15.6 from the finite prototype to the continuum by showing that the causal preorder $\preceq_{\text{caus}}$ on $\mathfrak{A}_\infty$ has a unique compatible Lorentzian metric $g_{\mu\nu}$ on $M^4 = M^4$ satisfying:

1. **Signature**: $g_{\mu\nu}$ has signature $(-, +, +, +)$ (from Theorem 15.6 lifted to the continuum);
2. **Causal compatibility**: the causal cone of $g_{\mu\nu}$ coincides with $\preceq_{\text{caus}}$ restricted to infinitesimally nearby points;
3. **Uniqueness**: $g_{\mu\nu}$ is unique up to a positive conformal factor (the Hawking–King–McCarthy theorem applies).

This upgrades the Lorentzian geometry layer from Open (frontier in Paper BE) to Cond (conditional on Phase-A coherence hypothesis inherited from the finite prototype).

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1 The finite prototype and its three ingredients

Theorem 1.1 (NFC Theorem 15.6 — Lorentzian Signature (finite prototype))**.** *Let U be a Phase-A region with local flow space $V = T \oplus S$ (temporal line T from the ledger clock form, spatial subspace $S = \ker \tau$). Assume spatial isotropy (the local near-symmetry group $G_{\tilde{\sigma}}$ acts transitively on the unit sphere of S) and an admissibly invariant nondegenerate quadratic form Q satisfying cone-null compatibility. Then Q has Lorentzian signature $(-, +, \ldots, +)$ with the negative direction aligned with T.*

Proof. NFC v8.41 Theorem 15.6 (Lemmas 15.3, 15.4, 15.5): spatial positivity, block diagonality, temporal sign forced negative by cone-null compatibility. □

Remark 1.2 (The three ingredients). Theorem 1.1 requires exactly three structures:

1. A distinguished irreversible direction T (the ledger clock form τ);
2. Spatial isotropy (the local near-symmetry group acts transitively on S); and
3. A causal cone (cone-null compatibility of the quadratic form Q).

All three are present in the continuum after Papers BC–BD.

2 The continuum versions of the three ingredients

Proposition 2.1 (Continuum ingredient 1: distinguished temporal direction)**.** *The temporal preorder $\preceq_{\mathrm{irr}}$ (Paper BC) provides a distinguished irreversible direction at every point of $\mathfrak{A}_\infty$: the ledger clock form τ is the differential of the compression count along continuation histories.*

Proposition 2.2 (Continuum ingredient 2: spatial isotropy)**.** *The gauge group $G = SU(3) \times SU(2) \times U(1)$ (Paper AV) acts on the spatial subspace $S = \ker \tau$ at each point via its fundamental representation. The spatial isotropy group at each point of $\mathfrak{A}_\infty$ contains the rotation group $SO(3) \subset SU(2)$, which acts transitively on the unit sphere of S for $\dim S = 3$.*

Proof. The $SU(2)$ factor of the gauge group G contains $SO(3)$ as a quotient, acting on the spatial directions via the adjoint representation. Since $\dim S = 3$ for the $\tau = 4$ sector (KPO-3 encoding, NFC monograph), and $SO(3)$ acts transitively on S^2, spatial isotropy holds. □

Proposition 2.3 (Continuum ingredient 3: causal cone)**.** *The Lieb–Robinson causal cone from Paper BD provides the causal cone structure: $x \preceq_{\mathrm{caus}} y$ iff y is in the Lieb–Robinson future cone of x at speed $v_{\max}$. The quadratic form $Q(v) = v_{\max}^2 dt^2 - d\mathbf{x}^2$ satisfies cone-null compatibility: $\partial \preceq_{\mathrm{caus}} = \{v : Q(v) = 0\}$.*

3 The Lorentzian representation theorem

Theorem 3.1 (Hawking–King–McCarthy Theorem, applied)**.** *A causal preorder $\preceq_{\mathrm{caus}}$ on a 4-dimensional topological manifold M^4 that is:*

1. *locally finite (finitely many causal predecessors/successors in any compact set),*
2. *consistent with a distinguished temporal direction (time-orientable),*
3. *locally compatible with a quadratic cone structure,*

uniquely determines a conformal Lorentzian structure $[g_{\mu\nu}]$ on M^4 (up to positive conformal factor).

Proof. Hawking–King–McCarthy (1976), Malament (1977): a causal order satisfying these conditions uniquely determines the conformal class of a Lorentzian metric. The exact metric (up to conformal factor) requires a volume form, which in the suite's setting is provided by the transport invariant I_{nl}. □

Theorem 3.2 (Lorentzian Geometry Theorem)**.** *Under the Phase-A coherence hypothesis inherited from NFC Theorem 15.6, the causal preorder $\preceq_{\text{caus}}$ on $\mathfrak{A}_\infty$ (Paper BD) determines a unique Lorentzian metric $g_{\mu\nu}$ on $M^4 = M^4$ satisfying:*

(a) ***Signature*** $(-,+,+,+)$*: from Proposition 1.1 lifted to the continuum via Propositions 2.1–2.3;*

(b) ***Causal compatibility****: the light cones of $g_{\mu\nu}$ coincide with the boundary $\partial \preceq_{\text{caus}}$ at each point;*

(c) ***Uniqueness up to conformal factor****: from the HKM theorem (Theorem 3.1).*

Proof. The three ingredients of Theorem 1.1 hold in the continuum by Propositions 2.1–2.3. Applying Theorem 1.1, the local quadratic form Q has signature $(-,+,+,+)$. Assembling Q across M^4 into a global metric tensor $g_{\mu\nu}$ is possible because $\mathfrak{A}_\infty$ is a locally compact Hausdorff space (from its C^*-algebraic structure) and the causal preorder is locally finite. The HKM theorem gives uniqueness. □

4 Status ledger

Label	Claim	Status
BG1	NFC Thm 15.6 (finite prototype Lorentzian signature)	UNC (monograph)
BG2	Three continuum ingredients verified	UNC
BG3	HKM theorem provides conformal Lorentzian structure	UNC (applies theorem)
BG4	Lorentzian metric $g_{\mu\nu}$ on M^4, signature $(-,+,+,+)$	**COND** *(Phase-A)*
BH	Einstein Field Equations from curvature identification	OPEN (next)

5 Conclusion

Paper BG lifts the NFC monograph's finite Lorentzian prototype (Theorem 15.6) to the continuum. The causal preorder $\preceq_{\text{caus}}$ from Paper BD, combined with the temporal direction from Paper BC and the spatial isotropy from the gauge group of Paper AV, satisfies the three conditions of Theorem 15.6. The Hawking–King–McCarthy theorem then provides the unique compatible Lorentzian metric $g_{\mu\nu}$ on M^4.

The Lorentzian geometry is not postulated — it is derived from the ledger direction, gauge isotropy, and Lieb–Robinson propagation speed. The remaining question is whether the curvature of $g_{\mu\nu}$ satisfies the Einstein Field Equations. That is Paper BH.